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UNIVERSITY INSTITUTE OF COMPUTING

CASE STUDY

ON

LifeOS – Personal Digital Command Center

OBJECT ORIENTED PROGRAMMING
25CAH-201

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CASE STUDY

LIFEOS – PERSONAL DIGITAL COMMAND CENTER

1. INTRODUCTION

LifeOS – Personal Digital Command Center is a desktop application developed using C++ and Qt. It is designed to help individuals manage their daily activities through a single platform.

In today's busy lifestyle, students and working professionals have to manage multiple responsibilities, including assignments, deadlines, personal expenses, daily habits, schedules, and important notes.

This case study explains how LifeOS can help solve these common organizational problems through a centralized desktop application.

2. BACKGROUND OF THE CASE

Consider a college student who has to manage several activities every day, such as:

- Completing college assignments.
- Remembering submission deadlines.
- Managing personal expenses.
- Maintaining daily study habits.
- Organizing class schedules.
- Saving important notes and ideas.

The student uses different applications, notebooks, and reminders to manage these activities. As a result, information becomes scattered, and it becomes difficult to maintain proper organization.

This situation creates a need for a single application that can manage different personal activities together.

3. PROBLEM IDENTIFICATION

The major problems identified in this situation are:

1. Difficulty in managing multiple daily tasks.
2. Forgetting important deadlines and activities.
3. Lack of proper expense tracking.
4. Difficulty maintaining daily habits.
5. Scattered notes and personal information.
6. Lack of an organized overview of daily productivity.

These problems can affect time management and personal productivity.

4. PROPOSED SOLUTION – LIFEOS

To address these problems, LifeOS was developed as a personal digital management application.

It provides different modules within one desktop application so that users can manage their activities from a centralized interface.

The application uses C++ for programming, Qt 6 for the graphical user interface, and SQLite for local data storage.

5. IMPLEMENTATION OF THE SOLUTION

LifeOS provides the following features to address the identified problems.

A. Dashboard

The Dashboard provides an overview of important information, including task completion, habits, expenses, and productivity.

B. Task Manager

The Task Manager allows users to add, edit, delete, search, and complete tasks. Users can also set priorities and due dates.

C. Calendar

The Calendar module is intended to help users organize important dates, events, and deadlines.

D. Habit Tracker

The Habit Tracker is designed to help users maintain daily routines and monitor their habits.

E. Personal Finance

The Personal Finance module allows users to record income and expenses and maintain financial transaction records.

F. Notes

The Notes module provides a place to store important information, ideas, and reminders.

G. Analytics

The Analytics module is designed to provide summarized information about activities such as productivity, tasks, habits, and finances.

6. WORKING OF THE APPLICATION

The general working process of LifeOS is:

1. The user opens the LifeOS desktop application.
2. The Dashboard displays the main interface.
3. The user selects the required module from the navigation panel.
4. The user enters or manages information in that module.
5. Supported information is stored in the local SQLite database.
6. The user can access the available information whenever required.

This process provides a centralized way to manage personal activities.

7. OBSERVED BENEFITS

The application offers several practical benefits:

- **Better Organization:** Different activities can be managed from one application.
- **Task Management:** Users can maintain a list of tasks and their completion status.
- **Financial Awareness:** Income and expenses can be recorded in one place.
- **Habit Management:** Users have a dedicated module for tracking routines.
- **Easy Access:** Different modules are accessible through a common interface.
- **Local Storage:** SQLite provides local storage for supported application data.

8. RESULT OF THE CASE STUDY

The case study demonstrates how a centralized desktop application can help reduce the difficulty of managing different personal activities.

LifeOS brings several personal management features together in one interface. It provides a structured way to organize tasks, maintain financial records, store notes, and access other available modules.

The application demonstrates a practical use of software development concepts to address everyday organizational challenges.

Note: These are the intended and demonstrated benefits of the application, not results from a formal user survey.

9. LIMITATIONS OF THE SOLUTION

Although LifeOS provides several useful features, it has some limitations:

1. It is currently designed for desktop use.
2. Data is stored locally rather than synchronized online.
3. Internet-based synchronization is not available.
4. Some features can be improved with additional development.
5. Advanced reminders and notifications can be added in future versions.

10. FUTURE SCOPE

LifeOS can be improved further by introducing:

- Cloud synchronization.
- User login and authentication.
- Automatic reminders and notifications.
- Advanced financial analysis.
- Detailed productivity reports.
- Mobile application integration.
- Data backup and export functionality.

11. CONCLUSION

This case study shows how LifeOS – Personal Digital Command Center can address common problems related to personal organization and activity management.

By combining task management, calendar, habit tracking, personal finance, notes, and analytics in one desktop application, LifeOS provides a centralized approach to managing everyday activities.

The development of this project also provided practical experience in C++, Qt 6, SQLite, database management, and graphical user interface development.

Overall, LifeOS demonstrates how technology can be used to make personal activity management more organized, accessible, and efficient.